

# Git & GitHub Student Training Guide

PES2UG23CS285

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## Step 1: Configure Git

```
● (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git config --list
credential.helper=osxkeychain
user.name=krishank-kureti
user.email=krishankkureti@gmail.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
core.ignorecase=true
core.precomposeunicode=true
remote.origin.url=https://github.com/krishank-kureti/my-calculator.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
branch.main.remote=origin
branch.main.merge=refs/heads/main
branch.main.vscode-merge-base=origin/main
```

## Step 3: Install Requirements

```
● (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % pip install -r requirements.txt
Requirement already satisfied: pytest in ./venv/lib/python3.13/site-packages (from -r requirements.txt (line 1)) (8.4.2)
Collecting pytest-cov (from -r requirements.txt (line 2))
  Downloading pytest_cov-7.0.0-py3-none-any.whl.metadata (31 kB)
Collecting pylint (from -r requirements.txt (line 3))
  Downloading pylint-3.3.8-py3-none-any.whl.metadata (12 kB)
Collecting click (from -r requirements.txt (line 4))
  Downloading click-8.3.0-py3-none-any.whl.metadata (2.6 kB)
Collecting requests (from -r requirements.txt (line 5))
  Using cached requests-2.32.5-py3-none-any.whl.metadata (4.9 kB)
Collecting bandit (from -r requirements.txt (line 6))
  Downloading bandit-1.8.6-py3-none-any.whl.metadata (6.9 kB)
Collecting black (from -r requirements.txt (line 7))
  Downloading black-25.9.0-cp313-cp313-macosx_11_0_arm64.whl.metadata (83 kB)
Requirement already satisfied: iniconfig>=1 in ./venv/lib/python3.13/site-packages (from pytest->-r requirements.txt (line 1)) (2.1.0)
Requirement already satisfied: packaging>=20 in ./venv/lib/python3.13/site-packages (from pytest->-r requirements.txt (line 1)) (25.0)
Requirement already satisfied: pluggy<2,>=1.5 in ./venv/lib/python3.13/site-packages (from pytest->-r requirements.txt (line 1)) (1.6.0)
Requirement already satisfied: pygments>=2.7.2 in ./venv/lib/python3.13/site-packages (from pytest->-r requirements.txt (line 1)) (2.19.2)
```

## Step 4: Run Tests

```
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % pytest tests/unit/ -v
===== test session starts =====
platform darwin -- Python 3.13.7, pytest-8.4.2, pluggy-1.6.0 -- /Users/krishankkureti/my-calculator/.venv/bin/python3.13
cachedir: .pytest_cache
rootdir: /Users/krishankkureti/my-calculator
configfile: pytest.ini
plugins: cov-7.0.0
collected 4 items

tests/unit/test_calculator.py::TestBasicOperations::test_add_positive_numbers PASSED [ 25%]
tests/unit/test_calculator.py::TestBasicOperations::test_subtract_positive_numbers PASSED [ 50%]
tests/unit/test_calculator.py::TestMultiplyDivideWithValidation::test_multiply_input_validation PASSED [ 75%]
tests/unit/test_calculator.py::TestMultiplyDivideWithValidation::test_divide_input_validation PASSED [100%]

===== 4 passed in 0.01s =====
```

## Step 5: Add New Test Cases

```
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % pytest tests/unit/ -v
===== test session starts =====
platform darwin -- Python 3.13.7, pytest-8.4.2, pluggy-1.6.0 -- /Users/krishankkureti/my-calculator/.venv/bin/python3.13
cachedir: .pytest_cache
rootdir: /Users/krishankkureti/my-calculator
configfile: pytest.ini
plugins: cov-7.0.0
collected 6 items

tests/unit/test_calculator.py::TestBasicOperations::test_add_positive_numbers PASSED [ 16%]
tests/unit/test_calculator.py::TestBasicOperations::test_subtract_positive_numbers PASSED [ 33%]
tests/unit/test_calculator.py::TestBasicOperations::test_add_negative_numbers PASSED [ 50%]
tests/unit/test_calculator.py::TestBasicOperations::test_subtract_negative_numbers PASSED [ 66%]
tests/unit/test_calculator.py::TestMultiplyDivideWithValidation::test_multiply_input_validation PASSED [ 83%]
tests/unit/test_calculator.py::TestMultiplyDivideWithValidation::test_divide_input_validation PASSED [100%]

===== 6 passed in 0.02s =====
```

## Step 6: Track & Push Changes

```
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git add .
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   tests/unit/test_calculator.py

• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git commit -m "Added negative number test cases"
[main 1f40972] Added negative number test cases
 1 file changed, 8 insertions(+)
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git push origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 559 bytes | 559.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/krishank-kureti/my-calculator.git
635e659..1f40972  main -> main
```

## Step 7: Create a Branch

```
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git branch
  feature/multi-div
* main
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git checkout feature/multi-div
Switched to branch 'feature/multi-div'
```

## Step 8: Add Multiply & Divide Functions

```
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % pytest tests/unit/ -v
===== test session starts =====
platform darwin -- Python 3.13.7, pytest-8.4.2, pluggy-1.6.0 -- /Users/krishankkureti/my-calculator/.venv/bin/python3.13
cachedir: .pytest_cache
rootdir: /Users/krishankkureti/my-calculator
configfile: pytest.ini
plugins: cov-7.0.0
collected 9 items

tests/unit/test_calculator.py::TestBasicOperations::test_add_positive_numbers PASSED [ 11%]
tests/unit/test_calculator.py::TestBasicOperations::test_subtract_positive_numbers PASSED [ 22%]
tests/unit/test_calculator.py::TestBasicOperations::test_add_negative_numbers PASSED [ 33%]
tests/unit/test_calculator.py::TestBasicOperations::test_subtract_negative_numbers PASSED [ 44%]
tests/unit/test_calculator.py::TestMultiplyDivideWithValidation::test_multiply_input_validation PASSED [ 55%]
tests/unit/test_calculator.py::TestMultiplyDivideWithValidation::test_divide_input_validation PASSED [ 66%]
tests/unit/test_calculator.py::TestMultiplyDivide::test_multiply_numbers PASSED [ 77%]
tests/unit/test_calculator.py::TestMultiplyDivide::test_divide_numbers PASSED [ 88%]
tests/unit/test_calculator.py::TestMultiplyDivide::test_divide_by_zero PASSED [100%]

===== 9 passed in 0.02s =====
```

## Step 9: Commit & Push Branch Changes

```
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git branch
* feature/multi-div
  main
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git add .
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git commit -m "Added Multiply and Divide"
[feature/multi-div 7cee866] Added Multiply and Divide
 1 file changed, 1 insertion(+), 9 deletions(-)
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git push origin feature/multi-div
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 470 bytes | 470.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
remote:
remote: Create a pull request for 'feature/multi-div' on GitHub by visiting:
remote:   https://github.com/krishank-kureti/my-calculator/pull/new/feature/multi-div
remote:
To https://github.com/krishank-kureti/my-calculator.git
 * [new branch]      feature/multi-div -> feature/multi-div
```

## Step 10: Merge Branch into Main

```
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git merge feature/multi-div
Updating 1f40972..7cee866
Fast-forward
 tests/unit/test_calculator.py | 10 +-----
 1 file changed, 1 insertion(+), 9 deletions(-)
```

## Step 11: Create Pull Request on GitHub

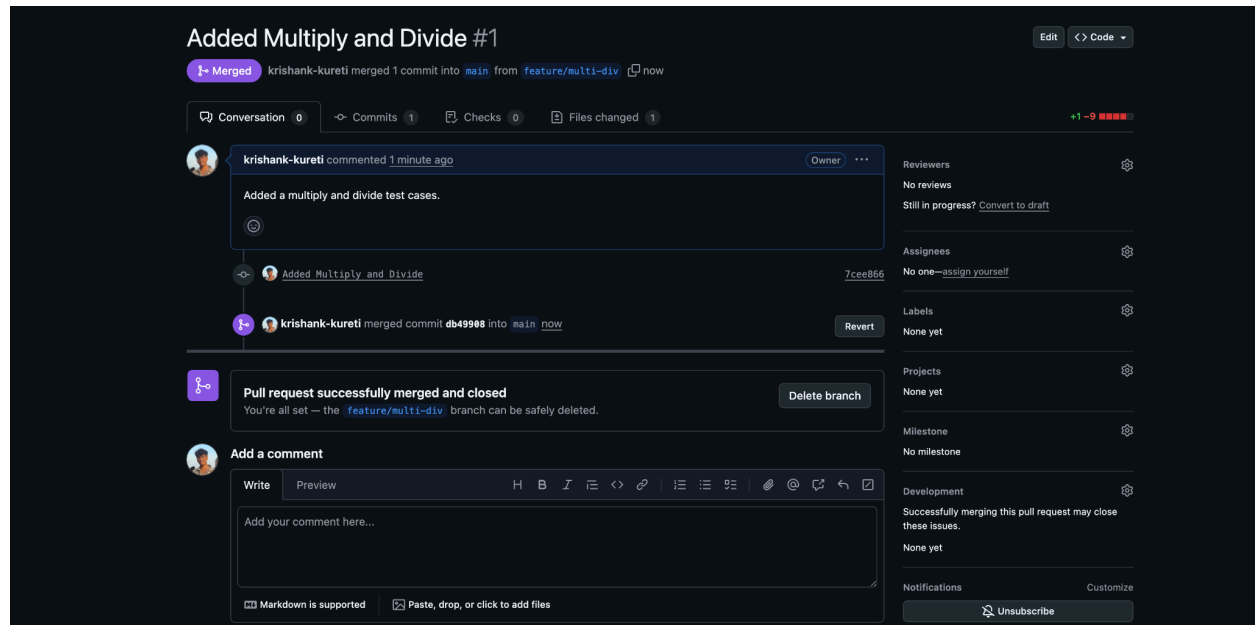
The screenshot shows a GitHub Pull Request (PR) titled "Added Multiply and Divide #1" in the repository "krishank-kureti / my-calculator". The PR is created by "krishank-kureti" and aims to merge 1 commit from the "feature/multi-div" branch into the "main" branch. The PR status is "Open".

The PR description includes a comment from "krishank-kureti" stating: "Added a multiply and divide test cases." Below this, a commit titled "Added Multiply and Divide" is shown with commit hash "7cee866".

A green checkmark indicates "No conflicts with base branch", stating "Merging can be performed automatically." A button labeled "Merge pull request" is visible, along with a link to "View command line instructions".

On the right side, the "Reviewers" section shows "No reviews". The "Assignees" section shows "No one—assign yourself". The "Labels" section shows "None yet". The "Projects" section shows "None yet". The "Milestone" section shows "No milestone". The "Development" section shows "Successfully merging this pull request may close these issues." and "None yet".

At the bottom, there is a "Close pull request" button and a "Comment" button. A notification bar at the bottom states: "You're receiving notifications because you authored the thread."



## Step 12: Sync Local with Remote

```

• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git fetch origin main
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 922 bytes | 922.00 KiB/s, done.
From https://github.com/krishank-kureti/my-calculator
 * branch      main      -> FETCH_HEAD
   1f40972..db49908  main      -> origin/main
• (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git pull
Updating 7cee866..db49908
Fast-forward

```

### Step 13: Delete Merged Branches

```

● (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git branch -d feature/multi-div
Deleted branch feature/multi-div (was 7cee866).
● (.venv) krishankkureti@Krishanks-MacBook-Air my-calculator % git push origin --delete feature/multi-div
To https://github.com/krishank-kureti/my-calculator.git
- [deleted]          feature/multi-div

```